



Smart India Hackathon - 2026

Internal Hackathon - Fergusson College (Autonomous)



SMART INDIA
HACKATHON
2026

- PS ID : SIH25074
- PS Title : **AI-Powered Personal Farming Assistant for Kerala Farmers**
- Theme : **Agriculture, FoodTech & Rural Development**
- PS Category : **Software**
- Team ID :
- Team Name : **ByteCraft**

Team Members:

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IDEA - Annadaata.AI



Annadaata.AI is WhatsApp-based AI Assistant which will enable farmers to get real-time crop recommendation, instant crop disease diagnosis and native-language voice support. This specific project currently focuses on farmers from Kerala.

Objectives:

1. **Bridge the digital literacy gap** by offering an intuitive, voice-enabled Malayalam AI assistant.
2. **Reduce crop losses through early detection** by scanning real-time plant photo using Computer Vision methods to identify crop diseases and recommend treatments.
3. **Provide microclimate-specific crop recommendations** leveraging Machine Learning models tailored to Kerala's varied terrain.
4. **Equipping rural farmers** with actionable, timely insights to increase crop productivity and **reduce agricultural financial distress**.





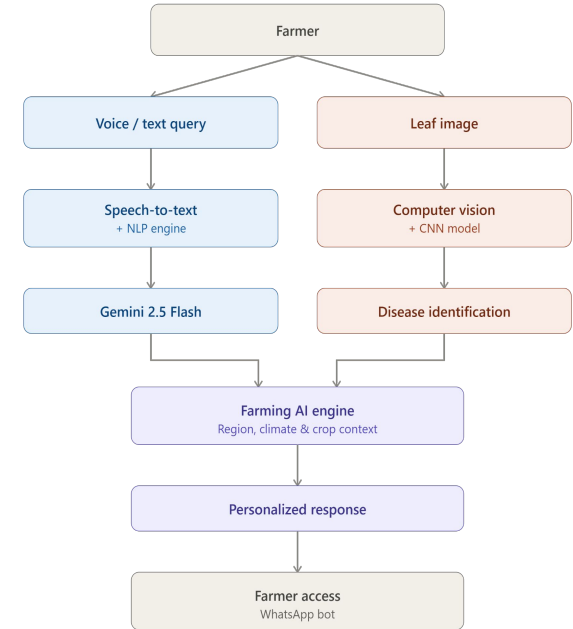
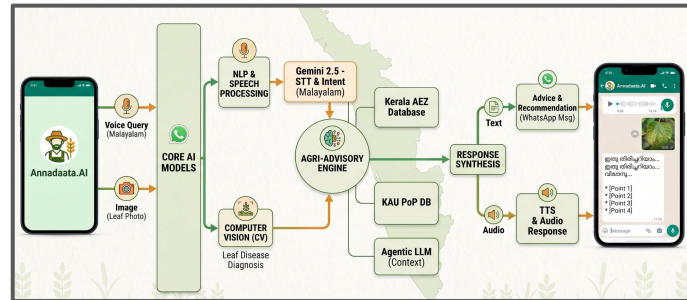
Technical Approach



The Technical approach will be as follows:

- WhatsApp API for the AI-assistant to work as a chatbot + voice assistant.
- NLP for voice/text query acceptance in Malayalam using Gemini Flash 2.5 model.
- CV based model (CNN) for real-time disease identification and diagnosis.
- ML model OR Agentic LLM model to get context of region and climate of the end user.
- Output as chat in WhatsApp regarding the query asked by end user.

Visual Representation of Annadaata.AI





Feasibility and Viability



KEY CHALLENGES & MITIGATION

1. **Low Digital Literacy** → WhatsApp-First Access enables familiar text, voice & image interaction with no new app to install.
2. **Regional Language & Dialects** → Malayalam-First AI with optimised prompts for regional farming terminology.
3. **Real-World Data Gaps** → Modular AI Architecture with independently testable services (crop advisory, disease detection, chat).
4. **Prediction Reliability** → Confidence-Based Assistance triggers expert referral instead of unreliable guidance.

VIABILITY & SCALABILITY

- Proven tech stack: Python, CNN, ML, WhatsApp Business API, Gemini 2.5 Flash
- Lightweight cloud architecture scales without redesign.
- Regional datasets expand progressively across Kerala.



Impact and Benefits

Aspects	Current Situation (Ground Reality)	Impact with Annadaata.ai
Disease Diagnosis	Farmers physically visit Krishi Bhavan, leading to delayed treatment.	Instant leaf scan via CNN gives real-time diagnostic remedies.
Expert Reach	Overburdened Krishi Bhavan officers struggle to cover every farm.	Serves as a 24/7 Virtual Assistant, triaging routine queries.
Digital Access	Complex portals and apps create high barriers for low-literacy farmers.	Zero-learning-curve WhatsApp voice interface in Malayalam.
Yield & Planning	Crop selection relies on guesswork despite microclimate shifts.	Data-driven ML crop recommendations tailored to local terrain.



Research and References



•Details / Links of the reference and research work

Government & Official Data

Department of Agriculture Development and Farmers' Welfare, Government of Kerala.

<https://ijarcs.info/index.php/ijarcs/article/view/7470>

Academic Studies & Reports

Sherin, P. S., & Haseena, V. A. (2023). Agriculture Distress and Farmers' Suicide in Kerala in the Globalization Era, IJAE; regional agro-climatic and pest analysis of Kerala farming.

Core Technologies

Meta WhatsApp Business API, Google Gemini 2.5 Flash LLM, PlantVillage CNN datasets.